

Real-Time FPGA CNN Hardware Accelerator IP (ops_convrelu)

Complete From-Scratch Design, Row-Stationary Line Buffers,
DSP48 Mapping, Timing Closure, and Interview Defense Guide

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Platform: AMD Xilinx Artix-7 (XC7A200T, Nexys Video)
EDA Tool: AMD Xilinx Vivado Design Suite

Abstract

This monograph details the complete from-scratch microarchitecture, arithmetic quantization, and physical timing closure of a standalone Convolutional Neural Network (CNN) hardware accelerator. Eliminating CPU software scheduling overhead and memory bandwidth saturation caused by standard ‘im2col’ GEMM expansions, this design pioneers a Row-Stationary line-buffer streaming memory hierarchy with fused 2D convolution and ReLU activation. Processing 16-bit signed Q1.5.10 fixed-point data across 120 spatially unrolled PE rows, the core utilizes 128-bit dual-port BRAM interfaces, explicit DSP48E1 pipelining directives, and glitch-free ‘BUFGCE’ synchronous clock gating, closing physical timing at **70 MHz (WNS = +0.092 ns)** with **83% DSP utilization** and **1.483 W total power**.

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Chapter 1

Stage 1: Specifications & Workload Analysis

1.1 Workload Bottlenecks in Edge CNN Inference

In embedded vision and edge classification workloads, 2D Convolution layers account for over 90% of total inference compute cycles. Standard software frameworks execute convolution by flattening feature maps using ‘im2col’ to leverage BLAS GEMM routines. For a 3×3 kernel, ‘im2col’ inflates memory storage and external memory bandwidth by $9\times$. On an embedded FPGA with limited DDR bandwidth, this causes severe memory stalls.

1.2 Design Targets

- **Input Resolution:** 28×28 grayscale feature maps (MNIST classification).
- **Arithmetic Format:** Signed 16-bit fixed-point (‘Q1.5.10’): 1 sign bit, 5 integer bits, 10 fractional bits.
- **Throughput Target:** Real-time streaming at 70 MHz, processing 8 pixels/cycle.
- **Accuracy Target:** $> 98\%$ test classification accuracy.

Chapter 2

Stage 2: Microarchitectural Innovations

2.1 Row-Stationary Line-Buffer Memory Hierarchy

To completely bypass ‘im2col’ memory expansion, a hardware line-buffer architecture was designed:

- Two BRAM-based FIFO line buffers (‘line_buf0’, ‘line_buf1’) *each store exactly one full horizontal row of feature maps (16 bits)*.
- As pixels from the third row stream sequentially from external memory, the line buffers output corresponding pixels from the preceding two rows.
- A 3×3 spatial kernel window is assembled and presented to the MAC array every clock cycle with **zero external memory duplication**.

2.2 16-Bit Q1.5.10 Arithmetic & DSP48E1 Pipelining

Products of two 16-bit ‘Q1.5.10’ operands generate 32-bit ‘Q2.10.20’ results.

- **Saturation Unit:** Slices bits ‘[25:10]’ to extract the 16-bit ‘Q1.5.10’ result. Overflow bits ‘[31:25]’ are inspected: if positive overflow occurs, saturate to ‘16’sh7FFF’; if negative overflow occurs, saturate to ‘16’sh8000’.
- **Fused ReLU:** If bit ‘[15]’ is high (negative), output ‘16’h0000’.
- **DSP Mapping:** Synthesis directives force mapping onto Artix-7 DSP48E1 slices:

```
1 (* use_dsp = "yes" *) reg signed [31:0] product;
```

2.3 Glitch-Free Synchronous Clock Gating (‘BUFGCE’)

Using discrete LUT logic (‘assign gated_clk = clken’) *introduces clock hazards, glitches, and severe clock tree skew*

Chapter 3

Stage 3: Physical Implementation & Timing Closure

3.1 Artix-7 XC7A200T Sign-Off Metrics

Table 3.1: Post-Route Physical Implementation Metrics on Artix-7 XC7A200T.

Resource	Used	Available	Utilization (%)
DSP48E1 Slices	612	740	82.7%
Block RAM (RAMB36E1)	48	365	13.2%
Slice LUTs	24,120	134,600	17.9%
Slice Registers (FF)	31,450	269,200	11.7%
Clock Frequency	70.0 MHz	—	WNS = +0.092 ns
Total On-Chip Power	1.483 W	—	Dynamic: 1.291 W, Static: 0.192 W
Classification Accuracy	98.2%	—	MNIST Test Set

Chapter 4

Stage 4: Interview Defense Dialogues

Q1: Why did you choose Q1.5.10 fixed-point arithmetic instead of standard INT8 or FP16?

Candidate Defense: FP16 floating-point arithmetic requires complex exponent alignment, mantissa normalization, and leading-zero detection logic that consumes massive FPGA LUT and DSP resources. Standard INT8 had insufficient dynamic range for our weight distributions, leading to clipping and an accuracy drop to 94.1%. Q1.5.10 provides 5 integer bits (range $[-32, +31.999]$) and 10 fractional bits (resolution ≈ 0.000976), preserving over 98.2% test accuracy while utilizing simple, deterministic integer DSP48E1 hardware.

Q2: How did you close physical timing at 70 MHz on an Artix-7 device?

Candidate Defense: The initial unpipelined critical path spanned from the DSP multiplier output, through a 9-operand spatial adder reduction tree, into the saturation clamping and ReLU logic in a single cycle ($T_{comb} > 16.5$ ns, causing timing failure). We closed timing by inserting two levels of pipeline retiming registers: one register layer immediately after the DSP product output (matching the DSP48 M_{REG} pipeline register), and a second pipeline register between the horizontal adder tree and the saturation unit. This brought combinational path delay down to 14.19 ns, successfully achieving WNS = +0.092 ns at 70 MHz ($T_{clk} = 14.28$ ns).